
Session 21 (AIML) - Optional Task

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Branch: Artificial Intelligence and Machine Learning (AIML)

Domain: AIML (Artificial Intelligence and Machine Learning)

Github Repo Link : https://github.com/vaishnavi-rathiii/mountreach.solutions.internship-aiml-tasks_assignments

Github File Link : https://github.com/vaishnavi-rathiii/mountreach.solutions.internship-aiml-tasks_assignments/tree/main/Session21/Session21%20OptionalTask

Dataset Used: – Kaggle - House_Price_Prediction

Mini Project - Complete Streamlit App

House Price Prediction Web App

Program Code :

```
1  '''
2  House Price Prediction Web Application
3
4  This Streamlit application predicts the selling price of a house
5  using a trained Linear Regression model.
6  '''
7
8  # -----
9  # Import Required Libraries
10 # -----
11
12 # Import Streamlit for building the web application
13 import streamlit as st
14
15 # Import Pandas for handling data
16 import pandas as pd
17
18 # Import Joblib for loading the saved machine learning model
19 import joblib
20
21 # -----
22 # Load Saved Model and Preprocessing Objects
23 # -----
24
25 # Load the trained Linear Regression model
26 model = joblib.load("LR_House_Price.pkl")
27
28 # Load the StandardScaler object
29 scaler = joblib.load("scaler.pkl")
30
31 # Load the encoded column names
32 encoded_columns = joblib.load("columns.pkl")
33
```

```

34 # -----
35 # Configure Streamlit Page
36 # -----
37
38 # page_title sets the browser tab title.
39 # layout="centered" keeps the application centered on the page.
40 st.set_page_config(
41     page_title="House Price Predictor",
42     layout="centered"
43 )
44
45 # -----
46 # Title and Description
47 # -----
48
49 st.title("🏠 House Price Prediction")
50
51 st.write("Enter the house details below to predict its selling price.")
52
53 st.divider()
54

```

```

55 # -----
56 # Numerical Input Fields
57 # -----
58
59 bedrooms = st.number_input(
60     "Bedrooms",
61     min_value=1,
62     max_value=10,
63     value=3
64 )
65
66 bathrooms = st.number_input(
67     "Bathrooms",
68     min_value=1.0,
69     max_value=10.0,
70     value=2.0
71 )
72
73 sqft_living = st.number_input(
74     "Living Area (sqft)",
75     min_value=300,
76     max_value=15000,
77     value=2000
78 )
79
80 sqft_lot = st.number_input(
81     "Lot Area (sqft)",
82     min_value=500,
83     max_value=1000000,
84     value=5000
85 )
86

```

```
87 floors = st.number_input(  
88     "Floors",  
89     min_value=1.0,  
90     max_value=5.0,  
91     value=1.0  
92 )  
93  
94 waterfront = st.number_input(  
95     "Waterfront (0 = No, 1 = Yes)",  
96     min_value=0,  
97     max_value=1,  
98     value=0  
99 )  
100  
101 view = st.number_input(  
102     "View Rating",  
103     min_value=0,  
104     max_value=4,  
105     value=0  
106 )  
107  
108 condition = st.number_input(  
109     "Condition",  
110     min_value=1,  
111     max_value=5,  
112     value=3  
113 )  
114  
115 sqft_above = st.number_input(  
116     "Sqft Above Ground",  
117     min_value=300,  
118     max_value=10000,  
119     value=1500  
120 )
```

```

121
122 sqft_basement = st.number_input(
123     "Sqft Basement",
124     min_value=0,
125     max_value=5000,
126     value=0
127 )
128
129 yr_built = st.number_input(
130     "Year Built",
131     min_value=1900,
132     max_value=2025,
133     value=2000
134 )
135
136 yr_renovated = st.number_input(
137     "Year Renovated (0 if Never)",
138     min_value=0,
139     max_value=2025,
140     value=0
141 )
142
143 year = st.number_input(
144     "Sale Year",
145     min_value=2014,
146     max_value=2025,
147     value=2014
148 )
149
150 month = st.number_input(
151     "Sale Month",
152     min_value=1,
153     max_value=12,
154     value=5
155 )
156
157 # -----
158 # Categorical Input Fields
159 # -----
160
161 # Text input is used to enter city name.
162 city = st.text_input("City")
163
164 # Text input is used to enter state zip.
165 statezip = st.text_input("State ZIP")
166
167 # Selectbox provides predefined options and prevents invalid inputs.
168 country = st.selectbox(
169     "Country",
170     ["USA"]
171 )
172
173 # -----
174 # Predict Button
175 # -----
176
177 predict_button = st.button("Predict House Price")
178

```

```

179 # -----
180 # Prediction
181 # -----
182
183 if predict_button:
184     try:
185         # Create DataFrame from user inputs
186         input_data = pd.DataFrame({
187             "bedrooms": [bedrooms],
188             "bathrooms": [bathrooms],
189             "sqft_living": [sqft_living],
190             "sqft_lot": [sqft_lot],
191             "floors": [floors],
192             "waterfront": [waterfront],
193             "view": [view],
194             "condition": [condition],
195             "sqft_above": [sqft_above],
196             "sqft_basement": [sqft_basement],
197             "yr_built": [yr_built],
198             "yr_renovated": [yr_renovated],
199             "city": [city],
200             "statezip": [statezip],
201             "country": [country],
202             "year": [year],
203             "month": [month]
204         })
205     except:
206         pass

```



```

207
208     # Perform One-Hot Encoding
209     input_data = pd.get_dummies(input_data)
210
211     # Align columns with training data
212     input_data = input_data.reindex(
213         columns=encoded_columns,
214         fill_value=0
215     )
216
217     # Numerical columns used for scaling
218     numerical_columns = [
219         "bedrooms",
220         "bathrooms",
221         "sqft_living",
222         "sqft_lot",
223         "floors",
224         "waterfront",
225         "view",
226         "condition",
227         "sqft_above",
228         "sqft_basement",
229         "yr_built",
230         "yr_renovated",
231         "year",
232         "month"
233     ]
234
235     # Apply Standard Scaling
236     input_data[numerical_columns] = scaler.transform(
237         input_data[numerical_columns]
238     )
239
240     # Predict house price
241     prediction = model.predict(input_data)
242
243     # Display predicted price
244     st.success(
245         f"🏠 Predicted House Price: ${prediction[0]:,.2f}"
246     )
247
248     except Exception as e:
249
250         st.error("An error occurred while making the prediction.")
251
252         st.exception(e)

```

Output Screenshot :

A screenshot of a web browser displaying a 'House Price Prediction' application. The browser's address bar shows 'localhost:8501'. The page has a dark theme. At the top, there is a 'Deploy' button. Below the title, a message says 'Enter the house details below to predict its selling price.' The form contains six input fields, each with a minus and plus button for adjustment:

- Bedrooms: 3
- Bathrooms: 2.00
- Living Area (sqft): 2000
- Lot Area (sqft): 5000
- Floors: 1.00

A second screenshot of the same web application, showing the continuation of the input fields. The fields include:

- Waterfront (0 = No, 1 = Yes): 0
- View Rating: 0
- Condition: 3
- Sqft Above Ground: 1500
- Sqft Basement: 0
- Year Built: 2000
- Year Renovated (0 if Never): 0

localhost:8501

Deploy

Sale Year
2014 - +

Sale Month
5 - +

City
Seattle

State ZIP
WA 98103

Country
USA

Predict House Price

Predicted House Price: \$512,884.26